Modelling Satirical Text Comprehension Using Cross Recurrence Analysis

Introduction

Satire can be defined as discursive critique involving irony, humour, and play (Simpson, 2003; Skalicky, 2025). Comprehending satire requires decoding surface meaning, recognising a clash with reality, and inferring the author's intent, making misinterpretation common (Kuiper, 1984; Pfaff & Gibbs, 1997; Simpson, 2003). Unlike verbal irony, little empirical work has linked reading behaviour of satirical text to comprehension outcomes (e.g., Skalicky, 2023). This difference may exist because satirical meaning accumulates across text, making traditional reading time measures poorly suited for capturing this distributed process (Skalicky, 2022, 2023). Therefore, the present study examines reading behaviour and comprehension of English satirical text using cross-recurrence quantification analysis (CRQA), a method for assessing stability in complex systems (Wallot, 2017). Although used only sparingly in reading research (Wallot, 2017; Wallot et al., 2014) and never for satire, CRQA may offer a more sensitive way to test whether reading behaviour of satirical text is linked to comprehension outcomes.

Method

Native English speakers in the United States (n = 88; Age Mean = 40.5, SD = 12.7; 57% female) were recruited via Prolific. Participants read the introduction and first two stories (1,560 words) from *Politically Correct Bedtime Stories* (Garner, 1994), which satirises political correctness through exaggerated language. The stories were presented using cumulative self-paced reading, where each word appears sequentially as participants press a button. Comprehension was assessed using 1–7 Likert ratings of the stories (interesting, funny, satirical) and author (joking, entertaining, serious). To control for variation in lexical knowledge, participants also completed the LexTALE vocabulary test (Mean = 88.45, SD = 10.06, possible scores = 0-100). Exploratory factor analysis reduced survey data into three factors: stories entertaining (funny and interesting), author intent (entertain and not serious), and satire recognition (stories satirical & author joking). Following suggested practice for RT data (Wallot, 2017), recurrence rate (RR, measured as 0-100% recurrence) was computed using the *crqa* package in R, reflecting how often reading time patterns repeated across the text within an empirically derived threshold and embedding dimension (in this case, 65ms threshold across six-word spans). RTs below 100 or above 2000ms were excluded (<1% of data). Three linear regression models tested whether RR and LexTALE scores were associated with the comprehension factors (RR is an aggregated value making multilevel models unsuitable).

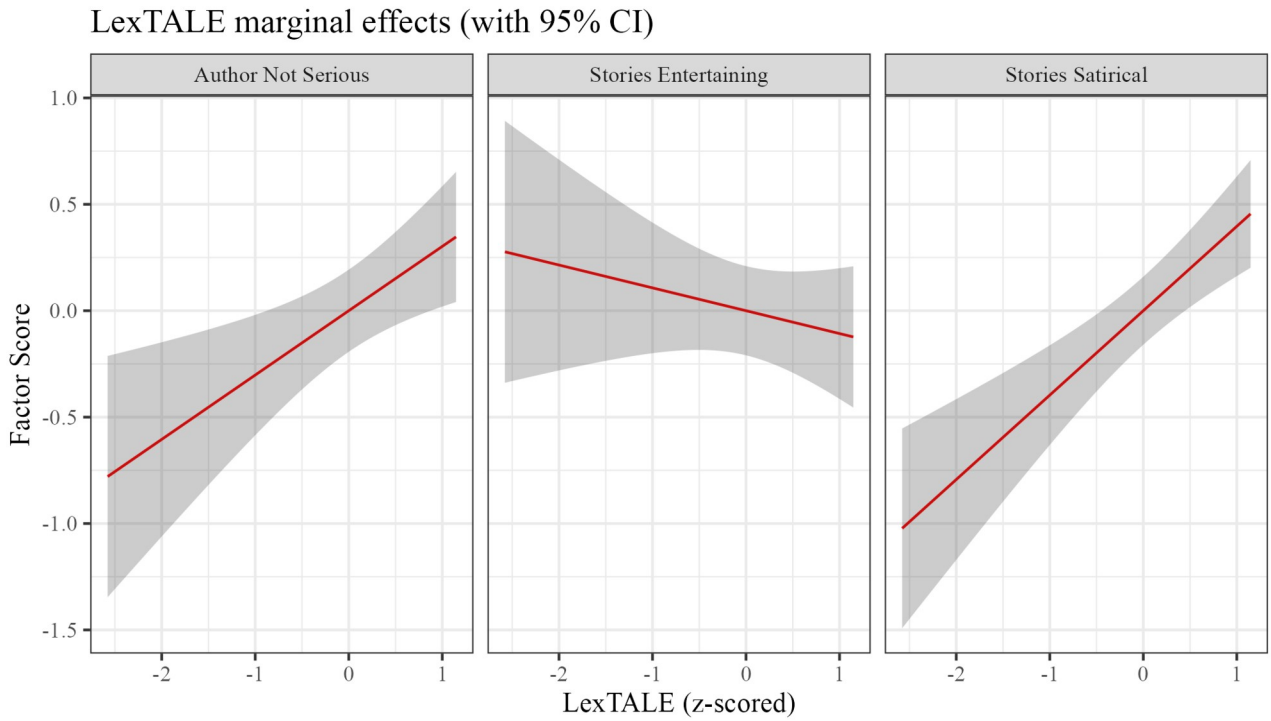
Results

RR values showed substantial variation among participants (M = 11.9%, SD = 12.2%, Range = 0.07% to 60.1%), indicating differences in stability of reading behaviour. However, regression analyses revealed that RR was not significantly associated with any comprehension outcome (all $p > 0.400$). In contrast, higher LexTALE scores significantly predicted recognition of author intent ($\beta = 0.302$, 95%CI[0.095, 0.509], $p < 0.001$, model adjusted $R^2 = 0.103$) and satire recognition ($\beta = 0.397$, 95%CI[0.226, 0.568], $p < 0.001$, model adjusted $R^2 = 0.227$), but not ratings of stories entertaining ($p = 0.334$). These effects are plotted in Figure 1.

Conclusion

This study explored whether inferences drawn from written satire may be better captured using methods from complex systems to capture stability in reading behaviour. In doing so, reading of satire was treated as a complex system when compared to standard approaches measuring reading time as a linear function of a text. Despite variation in reading behaviour stability for satirical texts, this stability was unrelated to comprehension outcomes. Instead, vocabulary knowledge was associated with detection of authorial intent and satirical meaning. These findings suggest that downstream inferences from satire depend more on reader background knowledge than on moment-to-moment reading dynamics. As such, these null findings align with prior research showing little to know influence of reading times on satire comprehension measures (Skalicky, 2022, 2023), suggesting satire comprehension operates independently of reading stability as measured using CRQA methods.

Figure 1. Effects of vocabulary knowledge on satire comprehension.



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